AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Sulfur dioxide (so2)
Observed / expected baseline	3.1 / -0.484828 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-19 16:00 UTC
Location	29.6240, -95.4743
Severity	severe
Detectors	zscore, stl, isolation_forest

The baseline is a mean of this series' own recent history, so it can fall below zero for a metric the network reports near or below its detection limit.

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 840	37.3 to 100; mean 74.9209	70.35 at 2026-07-19 15:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 840	22 to 37; mean 29.4469	30 at 2026-07-19 15:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 1120	0 to 320; mean 160.071	200 at 2026-07-19 15:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 1165	0 to 8.74555; mean 2.4283	2.05778 at 2026-07-19 15:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	11 / 132	5918.94 to 5946.98; mean 5930.45	5926.69 at 2026-07-19 18:00Z; 2 h after event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	11 / 132	64.8946 to 1860.78; mean 655.241	1356.63 at 2026-07-19 18:00Z; 2 h after event
noaa_gfs	Precipitable water (precipitable_water)	mm	11 / 132	29.224 to 47.063; mean 36.6002	36.0176 at 2026-07-19 18:00Z; 2 h after event
noaa_gfs	Surface pressure (surface_pressure)	hPa	11 / 132	1004.97 to 1018.53; mean 1012.58	1013.14 at 2026-07-19 18:00Z; 2 h after event
noaa_gfs	850-hPa temperature (t_850)	degC	11 / 132	17.9655 to 22.9209; mean 21.4415	22.0615 at 2026-07-19 18:00Z; 2 h after event
noaa_gfs	10-m eastward wind (u_10m)	m/s	11 / 132	-2.63161 to 2.90839; mean 0.7456	1.13125 at 2026-07-19 18:00Z; 2 h after event
noaa_gfs	10-m northward wind (v_10m)	m/s	11 / 132	-1.04667 to 4.7729; mean 2.1815	1.9878 at 2026-07-19 18:00Z; 2 h after event

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	Ozone (ozone)	ppm	16 / 1053	-0.004 to 0.08; mean 0.0182	0.018 at 2026-07-19 16:00Z; at event time
openaq	PM10 (pm10)	ug/m3	2 / 143	29 to 66; mean 41.7413	32 at 2026-07-19 16:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	76 / 3227	0.3 to 45.6; mean 7.7314	4.7 at 2026-07-19 16:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 357	0 to 100; mean 57.8739	92 at 2026-07-19 16:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 357	44 to 96; mean 76.1793	69 at 2026-07-19 16:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 357	0 to 0; mean 0	0 at 2026-07-19 16:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 357	1011 to 1020; mean 1015.38	1017 at 2026-07-19 16:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 357	24.26 to 37.22; mean 30.2348	31.89 at 2026-07-19 16:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 357	0 to 336; mean 195.541	186 at 2026-07-19 16:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 357	0.45 to 5.9; mean 2.0308	0.45 at 2026-07-19 16:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	72 / 5081	0 to 83.5; mean 9.2691	8.3 at 2026-07-19 16:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-19 19:42Z; 3 h 43 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-19 19:42Z; 3 h 43 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	4 / 4	0.0254212 to 0.0262849; mean 0.0259	0.0255937 at 2026-07-19 19:42Z; 3 h 43 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-19 19:42Z; 3 h 43 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	7 / 7	0.000157838 to 0.000213688; mean 0.0002	0.000176599 at 2026-07-19 20:09Z; 4 h 10 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-19 20:09Z; 4 h 10 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	4 / 4	1.64557e-05 to 5.83485e-05; mean 0	3.22877e-05 at 2026-07-19 20:09Z; 4 h 10 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-07-19 20:09Z; 4 h 10 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-19 19:42Z; 3 h 43 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	7 / 7	-3.60822e-05 to 0.000116366; mean 0	2.99088e-06 at 2026-07-19 20:09Z; 4 h 10 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-19 20:09Z; 4 h 10 min after event
tceq	Carbon monoxide (co)	ppm	3 / 182	0.1 to 0.8; mean 0.2187	0.3 at 2026-07-19 16:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Nitrogen dioxide (no2)	ppb	12 / 680	-2.5 to 20.7; mean 3.8793	0.8 at 2026-07-19 16:00Z; at event time
tceq	Sulfur dioxide (so2)	ppb	3 / 176	-0.6 to 6.9; mean 0.0915	3.1 at 2026-07-19 16:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-18 00Z 86.82, 06Z 90.86, 12Z 73.25, 18Z 58.09, 07-19 00Z 80.11, 06Z 93.72, 12Z 75.18, 18Z 54.34, 07-20 00Z 79.67, 06Z 94.44, 12Z 77.08, 18Z 48.96, 07-21 00Z 72.44
asos	Air temperature (temperature)	07-18 00Z 27.19, 06Z 26.48, 12Z 30.36, 18Z 33.1, 07-19 00Z 28.26, 06Z 25.26, 12Z 29.45, 18Z 33.97, 07-20 00Z 28.69, 06Z 24.95, 12Z 28.8, 18Z 34.38, 07-21 00Z 30.04
asos	Wind direction (wind_direction)	07-18 00Z 143.8, 06Z 120, 12Z 176, 18Z 182.5, 07-19 00Z 166.2, 06Z 81.77, 12Z 186.7, 18Z 139.9, 07-20 00Z 167.3, 06Z 133.3, 12Z 225.3, 18Z 176.1, 07-21 00Z 169.5
asos	Wind speed (wind_speed)	07-18 00Z 2.466, 06Z 1.905, 12Z 2.761, 18Z 3.809, 07-19 00Z 3.061, 06Z 0.926, 12Z 2.361, 18Z 2.239, 07-20 00Z 2.895, 06Z 1.554, 12Z 2.562, 18Z 2.234, 07-21 00Z 2.907
noaa_gfs	500-hPa geopotential height (gh_500)	07-18 06Z 5946, 12Z 5938, 18Z 5942, 07-19 00Z 5929, 06Z 5935, 12Z 5923, 18Z 5927, 07-20 00Z 5922, 06Z 5931, 12Z 5920, 18Z 5929, 07-21 00Z 5924
noaa_gfs	Planetary boundary layer height (pbl_height)	07-18 06Z 392.8, 12Z 268.8, 18Z 1591, 07-19 00Z 757.2, 06Z 284.4, 12Z 164.6, 18Z 1354, 07-20 00Z 691.3, 06Z 222.8, 12Z 142, 18Z 1254, 07-21 00Z 739.6
noaa_gfs	Precipitable water (precipitable_water)	07-18 06Z 39.17, 12Z 40.6, 18Z 43.53, 07-19 00Z 37.63, 06Z 36.82, 12Z 34.83, 18Z 36.18, 07-20 00Z 38.41, 06Z 33.19, 12Z 29.95, 18Z 32.96, 07-21 00Z 35.91
noaa_gfs	Surface pressure (surface_pressure)	07-18 06Z 1016, 12Z 1016, 18Z 1015, 07-19 00Z 1012, 06Z 1013, 12Z 1013, 18Z 1013, 07-20 00Z 1010, 06Z 1011, 12Z 1011, 18Z 1011, 07-21 00Z 1008
noaa_gfs	850-hPa temperature (t_850)	07-18 06Z 21.06, 12Z 20.78, 18Z 18.57, 07-19 00Z 20.78, 06Z 22.45, 12Z 21.95, 18Z 21.53, 07-20 00Z 20.98, 06Z 22.73, 12Z 22.67, 18Z 22.38, 07-21 00Z 21.42
noaa_gfs	10-m eastward wind (u_10m)	07-18 06Z -0.6589, 12Z 0.49, 18Z 0.4517, 07-19 00Z 0.153, 06Z 1.176, 12Z 0.6183, 18Z 1.386, 07-20 00Z 0.05235, 06Z 1.324, 12Z 1.67, 18Z 2.052, 07-21 00Z 0.2338
noaa_gfs	10-m northward wind (v_10m)	07-18 06Z 3.108, 12Z 2.039, 18Z 3.212, 07-19 00Z 3.971, 06Z 2.266, 12Z 1.697, 18Z 1.66, 07-20 00Z 2.493, 06Z 2.464, 12Z 0.6489, 18Z 0.8407, 07-21 00Z 1.78
openaq	Ozone (ozone)	07-18 00Z 0.01384, 06Z 0.01101, 12Z 0.01498, 18Z 0.02526, 07-19 00Z 0.01543, 06Z 0.008361, 12Z 0.01367, 18Z 0.03243, 07-20 00Z 0.01721, 06Z 0.006975, 12Z 0.01064, 18Z 0.04057, 07-21 00Z 0.02118
openaq	PM10 (pm10)	07-18 00Z 35, 06Z 34.36, 12Z 39.5, 18Z 37.18, 07-19 00Z 39.33, 06Z 35.58, 12Z 35.75, 18Z 46.83, 07-20 00Z 45.25, 06Z 44.75, 12Z 49.67, 18Z 50.75, 07-21 00Z 43.67
openaq	PM2.5 (pm25)	07-18 00Z 8.073, 06Z 8.918, 12Z 8.387, 18Z 5.677, 07-19 00Z 7.423, 06Z 8.501, 12Z 6.148, 18Z 8.112, 07-20 00Z 7.621, 06Z 8.657, 12Z 7.838, 18Z 6.568, 07-21 00Z 6.504
openweather	Cloud cover (cloud_cover)	07-18 00Z 85.5, 06Z 96.72, 12Z 84.16, 18Z 94.1, 07-19 00Z 94.31, 06Z 93.58, 12Z 94.52, 18Z 84.9, 07-20 00Z 12.83, 06Z 5.645, 12Z 3.833, 18Z 2.967, 07-21 00Z 4.15
openweather	Relative humidity (humidity)	07-18 00Z 84.1, 06Z 89.38, 12Z 79.68, 18Z 63.13, 07-19 00Z 77.14, 06Z 90.48, 12Z 81.63, 18Z 60.1, 07-20 00Z 74.79, 06Z 90.77, 12Z 80.67, 18Z 53.33, 07-21 00Z 67.35
openweather	Precipitation rate (precipitation)	07-18 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-19 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-20 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-21 00Z 0
openweather	Surface pressure (pressure)	07-18 00Z 1019, 06Z 1018, 12Z 1019, 18Z 1016, 07-19 00Z 1016, 06Z 1016, 12Z 1017, 18Z 1014, 07-20 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1012, 07-21 00Z 1012
openweather	Air temperature (temperature)	07-18 00Z 27.97, 06Z 26.98, 12Z 29.79, 18Z 34.26, 07-19 00Z 29.62, 06Z 26.2, 12Z 28.9, 18Z 35.34, 07-20 00Z 30.62, 06Z 25.88, 12Z 28.74, 18Z 35.79, 07-21 00Z 32.12
openweather	Wind direction (wind_direction)	07-18 00Z 182.3, 06Z 180.2, 12Z 194.5, 18Z 174.5, 07-19 00Z 183.4, 06Z 192.7, 12Z 213.6, 18Z 184.9, 07-20 00Z 186.7, 06Z 194.4, 12Z 239.7, 18Z 215.6, 07-21 00Z 189.6

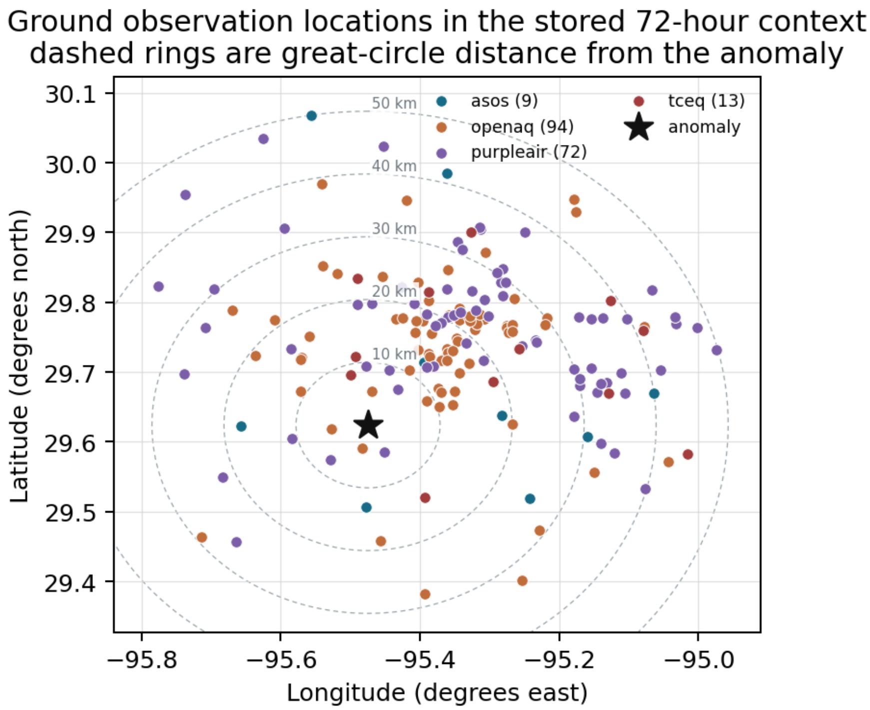
Source	Metric	Values
openweather	Wind speed (wind_speed)	07-18 00Z 1.868, 06Z 1.972, 12Z 2.012, 18Z 1.928, 07-19 00Z 2.57, 06Z 1.94, 12Z 1.676, 18Z 1.968, 07-20 00Z 2.389, 06Z 1.895, 12Z 1.845, 18Z 1.766, 07-21 00Z 2.68
purpleair	PM2.5 (pm25)	07-18 00Z 9.86, 06Z 10.92, 12Z 9.609, 18Z 7.344, 07-19 00Z 8.974, 06Z 10.5, 12Z 8.618, 18Z 8.636, 07-20 00Z 9.265, 06Z 10.82, 12Z 9.817, 18Z 7.625, 07-21 00Z 8.797
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	CO column (s5p_co_column)	07-19 18Z 0.02551, 07-20 18Z 0.02628
sentinel5p	CO granules available (s5p_co_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-18 18Z 0.0001745, 07-19 18Z 0.0001766, 07-20 18Z 0.000213
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-18 18Z 1.872e-05, 07-19 18Z 3.229e-05, 07-20 18Z 5.835e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-18 18Z 3.278e-05, 07-19 18Z 2.991e-06, 07-20 18Z 1.73e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-18 18Z 1, 07-19 18Z 1, 07-20 18Z 1
tceq	Carbon monoxide (co)	07-18 06Z 0.2111, 12Z 0.2556, 18Z 0.2556, 07-19 00Z 0.25, 06Z 0.1944, 12Z 0.2111, 18Z 0.2, 07-20 00Z 0.2667, 06Z 0.2056, 12Z 0.2125, 18Z 0.1722, 07-21 00Z 0.2222
tceq	Nitrogen dioxide (no2)	07-18 06Z 3.628, 12Z 3.301, 18Z 2.128, 07-19 00Z 1.933, 06Z 3.03, 12Z 2.541, 18Z 3.297, 07-20 00Z 3.545, 06Z 4.577, 12Z 6.21, 18Z 6.136, 07-21 00Z 6.723
tceq	Sulfur dioxide (so2)	07-18 06Z -0.3611, 12Z -0.3222, 18Z -0.2889, 07-19 00Z -0.1, 06Z -0.4533, 12Z 0.1056, 18Z 0.4389, 07-20 00Z -0.2875, 06Z -0.3471, 12Z -0.175, 18Z 1.906, 07-21 00Z 1.213

Per-site averages

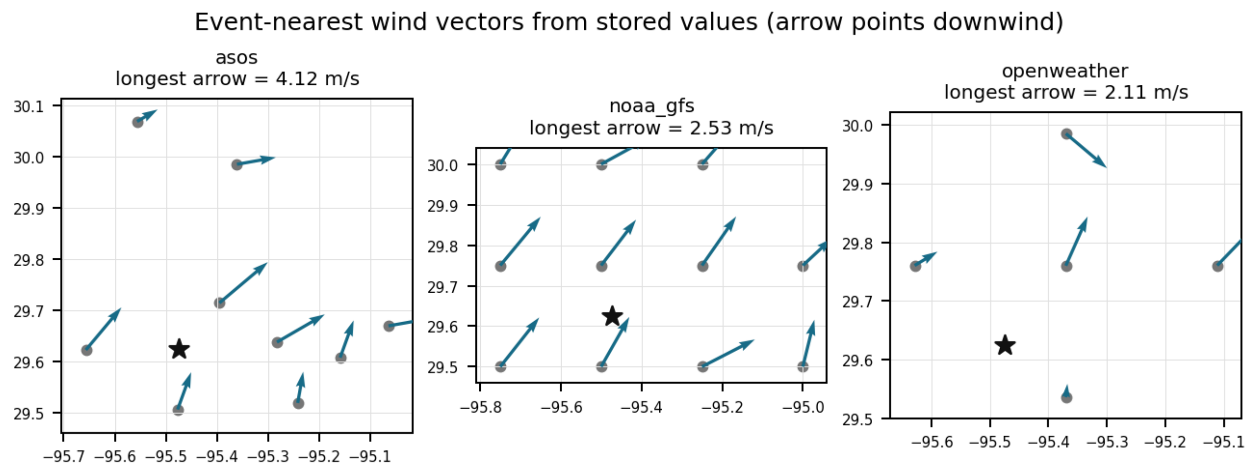
Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	Relative humidity (humidity)	MCJ (12.608 km) 72.43, AXH (13.11 km) 72.48, SGR (17.608 km) 79.95, HOU (18.613 km) 74.61, LVJ (25.353 km) 75.27, EFD (30.569 km) 74.76, T41 (39.963 km) 74.99, IAH (41.55 km) 73.28, +1 more
asos	Air temperature (temperature)	MCJ (12.608 km) 29.83, AXH (13.11 km) 29.39, SGR (17.608 km) 28.71, HOU (18.613 km) 29.7, LVJ (25.353 km) 29.32, EFD (30.569 km) 29.86, T41 (39.963 km) 29.84, IAH (41.55 km) 29.99, +1 more
asos	Wind direction (wind_direction)	MCJ (12.608 km) 170.6, AXH (13.11 km) 103.9, SGR (17.608 km) 173.3, HOU (18.613 km) 192.6, LVJ (25.353 km) 134.3, EFD (30.569 km) 169, T41 (39.963 km) 185, IAH (41.55 km) 167.9, +1 more
asos	Wind speed (wind_speed)	MCJ (12.608 km) 3.252, AXH (13.11 km) 0.9825, SGR (17.608 km) 2.726, HOU (18.613 km) 3.155, LVJ (25.353 km) 1.786, EFD (30.569 km) 3.06, T41 (39.963 km) 3.31, IAH (41.55 km) 2.244, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.50,-95.50 (14.007 km) 5931, gfs:29.75,-95.50 (14.23 km) 5930, gfs:29.50,-95.25 (25.708 km) 5931, gfs:29.75,-95.25 (25.808 km) 5931, gfs:29.50,-95.75 (30.014 km) 5931, gfs:29.75,-95.75 (30.09 km) 5930, gfs:30.00,-95.50 (41.885 km) 5930, gfs:30.00,-95.25 (47.082 km) 5930, +3 more

Source	Metric	Values
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.50,-95.50 (14.007 km) 499.5, gfs:29.75,-95.50 (14.23 km) 903, gfs:29.50,-95.25 (25.708 km) 480.1, gfs:29.75,-95.25 (25.808 km) 789.9, gfs:29.50,-95.75 (30.014 km) 465.6, gfs:29.75,-95.75 (30.09 km) 581.5, gfs:30.00,-95.50 (41.885 km) 902.2, gfs:30.00,-95.25 (47.082 km) 864.3, +3 more
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.50,-95.50 (14.007 km) 36.77, gfs:29.75,-95.50 (14.23 km) 36.37, gfs:29.50,-95.25 (25.708 km) 37.29, gfs:29.75,-95.25 (25.808 km) 37.19, gfs:29.50,-95.75 (30.014 km) 35.88, gfs:29.75,-95.75 (30.09 km) 35.44, gfs:30.00,-95.50 (41.885 km) 36, gfs:30.00,-95.25 (47.082 km) 36.89, +3 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.50,-95.50 (14.007 km) 1013, gfs:29.75,-95.50 (14.23 km) 1012, gfs:29.50,-95.25 (25.708 km) 1014, gfs:29.75,-95.25 (25.808 km) 1014, gfs:29.50,-95.75 (30.014 km) 1012, gfs:29.75,-95.75 (30.09 km) 1011, gfs:30.00,-95.50 (41.885 km) 1011, gfs:30.00,-95.25 (47.082 km) 1012, +3 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.50,-95.50 (14.007 km) 21.43, gfs:29.75,-95.50 (14.23 km) 21.2, gfs:29.50,-95.25 (25.708 km) 21.56, gfs:29.75,-95.25 (25.808 km) 21.41, gfs:29.50,-95.75 (30.014 km) 21.53, gfs:29.75,-95.75 (30.09 km) 21.51, gfs:30.00,-95.50 (41.885 km) 21.23, gfs:30.00,-95.25 (47.082 km) 21.12, +3 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.50,-95.50 (14.007 km) 0.4518, gfs:29.75,-95.50 (14.23 km) 0.9068, gfs:29.50,-95.25 (25.708 km) 0.491, gfs:29.75,-95.25 (25.808 km) 0.6793, gfs:29.50,-95.75 (30.014 km) 0.9568, gfs:29.75,-95.75 (30.09 km) 1.023, gfs:30.00,-95.50 (41.885 km) 1.325, gfs:30.00,-95.25 (47.082 km) 0.4668, +3 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.50,-95.50 (14.007 km) 2.421, gfs:29.75,-95.50 (14.23 km) 2.099, gfs:29.50,-95.25 (25.708 km) 2.409, gfs:29.75,-95.25 (25.808 km) 2.203, gfs:29.50,-95.75 (30.014 km) 2.107, gfs:29.75,-95.75 (30.09 km) 2.129, gfs:30.00,-95.50 (41.885 km) 1.678, gfs:30.00,-95.25 (47.082 km) 1.708, +3 more
openaq	Ozone (ozone)	312 (0.017 km) 0.01762, 311 (8.339 km) 0.01837, 3055 (13.984 km) 0.01435, 1319333 (18.217 km) 0.02245, 325 (18.693 km) 0.004938, 287 (19.112 km) 0.01876, 320 (20.024 km) 0.01554, 319 (23.419 km) 0.01796, +8 more
openaq	PM10 (pm10)	15852419 (22.104 km) 36.51, 271 (33.811 km) 47.2
openaq	PM2.5 (pm25)	13955814 (3.744 km) 4.703, 15399615 (5.167 km) 4.866, 8967119 (5.456 km) 11.14, 14140752 (8.986 km) 8.615, 8967021 (10.262 km) 9.59, 8967001 (10.484 km) 6.46, 13955816 (10.805 km) 9.451, 15539683 (11.385 km) 8.711, +68 more
openweather	Cloud cover (cloud_cover)	grid:south (14.084 km) 57.74, grid:center (18.176 km) 56.39, grid:west (21.256 km) 56.19, grid:east (38.226 km) 58.76, grid:north (41.392 km) 60.31
openweather	Relative humidity (humidity)	grid:south (14.084 km) 75.81, grid:center (18.176 km) 74.63, grid:west (21.256 km) 75.35, grid:east (38.226 km) 80.32, grid:north (41.392 km) 74.8
openweather	Precipitation rate (precipitation)	grid:south (14.084 km) 0, grid:center (18.176 km) 0, grid:west (21.256 km) 0, grid:east (38.226 km) 0, grid:north (41.392 km) 0
openweather	Surface pressure (pressure)	grid:south (14.084 km) 1015, grid:center (18.176 km) 1015, grid:west (21.256 km) 1015, grid:east (38.226 km) 1015, grid:north (41.392 km) 1015
openweather	Air temperature (temperature)	grid:south (14.084 km) 30, grid:center (18.176 km) 30.74, grid:west (21.256 km) 30.22, grid:east (38.226 km) 29.9, grid:north (41.392 km) 30.32
openweather	Wind direction (wind_direction)	grid:south (14.084 km) 197.8, grid:center (18.176 km) 211.6, grid:west (21.256 km) 190.1, grid:east (38.226 km) 198.7, grid:north (41.392 km) 179.6
openweather	Wind speed (wind_speed)	grid:south (14.084 km) 2.328, grid:center (18.176 km) 1.783, grid:west (21.256 km) 1.623, grid:east (38.226 km) 2.636, grid:north (41.392 km) 1.785
purpleair	PM2.5 (pm25)	161003 (4.856 km) 5.355, 166451 (7.01 km) 10.41, 161015 (7.666 km) 10.68, 46237 (9.249 km) 0.4781, 186315 (9.424 km) 10.18, 276512 (10.819 km) 10.9, 293735 (12.303 km) 11.23, 293725 (13.024 km) 10.66, +64 more
tceq	Carbon monoxide (co)	482011052 (22.762 km) 0.3651, 482011035 (24.236 km) 0.1322, 482011039 (33.805 km) 0.15
tceq	Nitrogen dioxide (no2)	482010055 (8.334 km) 2.154, 482011066 (11.0 km) 4.546, 480391004 (13.97 km) 1.677, 482010416 (18.688 km) 4.636, 482011052 (22.762 km) 7.646, 482010047 (23.42 km) 3.771, 482011035 (24.236 km) 5.603, 482011039 (33.805 km) 2.633, +4 more
tceq	Sulfur dioxide (so2)	482010051 (0.0 km) 0.2767, 482011035 (24.236 km) -0.2534, 482011039 (33.805 km) 0.2448



Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly
50 km
40 km
30 km
20 km
10 km
asos (9)
openaq (94)
purpleair (72)
tceq (13)
anomaly
Latitude (degrees north)
Longitude (degrees east)
30.1
30.0
29.9
29.8
29.7
29.6
29.5
29.4
−95.8
−95.6
−95.4
−95.2
−95.0



Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 4.12 m/s
noaa_gfs
longest arrow = 2.53 m/s
openweather
longest arrow = 2.11 m/s

Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 35

The air quality anomaly in Houston, Texas is not caused by changes in temperature.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 35

The PM2.5 levels are elevated at 8.3 ug/m3, which is higher than the mean of 9.2691 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 35

The weather conditions on July 19th, including high pressure and light winds, contributed to the air quality anomaly in Houston, Texas.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 35

The pollutant showing the anomaly is sulfur dioxide measured by a TCEQ monitor.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 35

A monitor-specific artifact remains a viable alternative because the provided SO2 network includes only three TCEQ monitors within 50 km and the two other SO2 monitors have different site means and no explicit simultaneous co-spike reported at 2026-07-19 16:00 UTC, so independent spatial corroboration for the exact event is limited in the current dataset.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 35

A monitor-specific measurement artifact or ultra-local nonregional SO_2 source remains a plausible alternative explanation because Sentinel-5P did not show a matching broad SO_2 column enhancement over the area later that afternoon.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 35

Boundary layer dynamics and weak atmospheric ventilation limited the dispersion of emitted sulfur dioxide, leading to localized ground-level accumulation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 35

A broad regional sulfur event is contradicted because Sentinel-5P showed a low SO_2 column near the area later on 2026-07-19 at about 2.99e-06 mol/m^2, while satellite CO and HCHO were not elevated and surface PM2.5 near the anomaly time remained modest at about 4.7 to 8.3 ug/m3, which is not the signature of a large area-wide combustion or volcanic sulfur episode.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 35

Sentinel-5P measured an SO2 column density of 2.99 x 10^-6 mol/m^2 near the anomaly site approximately four hours after the surface spike at 20:09:34 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 35

Ground monitoring recorded a sharp SO2 concentration spike of 3.1 ppb at station 482010051 at 16:00 UTC on July 19, 2026, while Sentinel-5P column observations showed baseline atmospheric SO2 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 35

Low surface wind speeds and southerly wind directions facilitated the transport of SO2 plumes from upwind industrial sources toward the TCEQ monitoring location.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 35

Surface meteorology near the anomaly time showed weak southerly to south-southwesterly flow, with nearby observations around 0.45 to 2.06 m/s and wind directions near 186 to 200 degrees, which would favor transport of a nearby sulfur plume to the monitor with only modest dispersion.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.45 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.03549610439032991 m/s; n=357 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 35

A severe sulfur dioxide (SO_2) anomaly occurred on July 19, 2026, at 16:00 UTC at coordinates (29.624, -95.474), reaching an observed concentration of 3.1 ppb compared to an expected value of -0.485 ppb, resulting in a z-score of 30.03.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 35

A short-duration plume from nearby industrial sulfur-emitting sources such as refinery, petrochemical, sulfur recovery, storage, or ship-channel operations is the leading physical cause of the 3.1 ppb SO_2 spike because the anomaly was highly localized at one TCEQ monitor while concurrent NO_2, CO, PM2.5, PM10, and ozone near the site were not comparably elevated.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 35

The air quality anomaly in Houston, Texas on July 19th was characterized by elevated levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 35

The temperature inversion over the Houston area is causing a buildup of pollutants near the surface.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 35

Surface wind speeds near the Houston anomaly location averaged approximately 2 m/s from the south to south-southwest during the spike event on July 19, 2026.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.45 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.03549610439032991 m/s; n=357 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 35

A ground-level SO2 spike reaching 3.1 ppb was recorded at TCEQ monitor 482010051 in Houston at 16:00 UTC on July 19, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 35

The wind direction and speed are not effectively dispersing pollutants, leading to a concentration of air pollutants in the Houston area.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 35

A nearby industrial facility or power plant is releasing excessive amounts of particulate matter and volatile organic compounds into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 35

Weak southerly to south-southwesterly low-level flow and only moderate surface wind speeds likely increased the probability that a nearby emission plume intersected the monitor without rapid dilution at 2026-07-19 16:00 UTC.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.45 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.03549610439032991 m/s; n=357 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 35

The TCEQ monitor at 29.624, -95.474 measured 3.1 ppb SO_2 at 2026-07-19 16:00 UTC, which was far above the expected value of about -0.485 ppb and had a severe z-score of about 30, indicating an extreme local SO_2 excursion at that site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 35

The air quality anomaly in Houston, Texas is not caused by wind direction or speed.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 35

NO2 levels were also elevated during the air quality anomaly in Houston, Texas on July 19th.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 35

The anomalous sulfur dioxide measurement occurred at TCEQ monitor 482010051 in the Houston area, where the monitor mean over the context window was 0.2767 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 35

ASOS weather stations and OpenWeather reported light surface winds around 2 m/s from the south-southwest near the time of the SO2 anomaly, indicating local plume dispersion from upwind industrial facilities.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 35

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 35

Industrial point-source emissions or process upset events at upwind petrochemical facilities near Houston likely caused the elevated SO2 concentration detected by the ground station.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 35

NOAA GFS model output showed planetary boundary layer height expanding to over 1350 meters by 18:00 UTC on July 19, 2026, facilitating vertical dilution following early morning accumulation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 35

The SO2 concentration spike was localized to TCEQ monitor 482010051 at the anomaly location, whereas monitors 24.2 km and 33.8 km away maintained period averages of -0.2534 ppb and 0.2448 ppb, respectively.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 35

The TCEQ sulfur dioxide concentration at 29.624, -95.474 was 3.1 ppb at 2026-07-19T16:00:00+00:00, compared with an expected value of -0.48482758620689664 ppb and a z-score of 30.02768750590938, which is labeled severe.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 35

A local industrial sulfur plume is supported because the TCEQ SO2 monitor at the anomaly site measured 3.1 ppb at 2026-07-19 16:00 UTC against an expected value near -0.48 ppb and a monitor mean of 0.28 ppb, while nearby surface winds were weak and southerly to south-southwesterly at about 0.45 to 2.06 m/s, conditions consistent with limited dilution and transport from the Houston Ship Channel or other industrial sources to the north-northeast of air parcels arriving from the south.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 35

The anomaly occurred on July 19th at approximately 16:01:53 UTC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 35

Regional supporting data did not show a broad sulfur pollution episode, because Sentinel-5P SO2 near the area later that day was low, nearby NO2 and CO were not elevated at the anomaly time, and nearby PM2.5 and ozone remained modest, so the most likely cause is a localized industrial sulfur release rather than region-wide accumulation or wildfire smoke.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.